

St. Francis Institute of Technology, Mumbai-400 103
Department Of Information Technology

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Class: BE-ITA/B, Semester: VII

Subject: Healthcare Lab

Experiment 5

1. **Aim:** To extract Natural language Entity from medical reports using spacy.
2. **Objectives:** Students should be able to develop healthcare applications using appropriate computational tools.
3. **Prerequisite:** Python basics, A basic understanding of the goals and terminology of Natural Language Processing
4. **Pre-Experiment Exercise:**

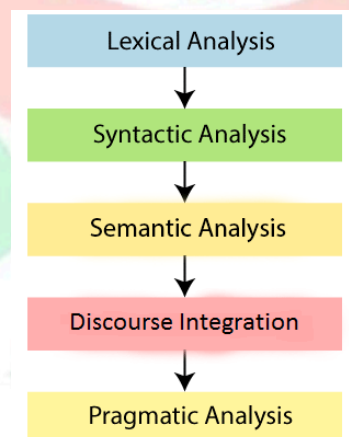
Theory:

Natural Language Processing:

Natural language processing (commonly referred to as NLP) is a subset of Artificial Intelligence research, which is concerned with machine learning modeling tasks, aimed at giving computer programs the ability to understand human language, both written and spoken.

Natural language processing is not only concerned with processing, as recent developments in the field such as the introduction of Large Language Models (LLMs) and GPT3, are also aimed at language generation as well.

Phases of NLP:



libraries to perform Clinical NLP:

- **spaCy:** NLP library that provides text processing and orchestration.
- **scispaCy:** a library of clinical and biomedical specific components that integrate with spaCy.
- **negspaCy:** for detecting negation of terms.

Pipelines:

Central to spaCy is the idea of a processing pipeline, the organized steps to extract information from a text. These steps consist of specific components chained together that act on a text to extract structured data.

The pipeline generates a Document or Doc object containing the structured data extracted from the text.

5. Laboratory Exercise**Procedure**

- i. Import spacy and scispacy library for NLP of healthcare data

```
2] 15s !python -m pip install --user spacy
!python -m spacy download en_core_web_sm

Requirement already satisfied: cloudpathlib<0.10.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.1.0->spacy) (0.7.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.1.0->spacy) (7.3.1)
Requirement already satisfied: MarkupSafe<2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: marisa-trie<1.1.0 in /usr/local/lib/python3.12/dist-packages (from language-data>1.2->langcodes<4.0.0,>=3.2.0->spacy) (1.3.1)
Requirement already satisfied: markdown-it-py<2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>10.11.0->typer<1.0.0,>=0.3.0->spacy) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>10.11.0->typer<1.0.0,>=0.3.0->spacy) (2.19.2)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.1.0->spacy) (1.17.3)
Requirement already satisfied: mdurl<0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py<2.2.0->rich>10.11.0->typer<1.0.0,>=0.3.0->spacy) (0.1.2)
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
    12.8/12.8 MB 46.8 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.
```

- ii. Extract information with trained pipelines.

```
import pandas as pd
raw_df = pd.read_csv("/content/mtsamples - mtsamples.csv", index_col=0)
#raw_df = pd.read_excel("C://dataset/mtsamples.xlsx", index_col=0)
raw_df.head()
```

	description	medical_specialty	sample_name	transcription	keywords
0	A 23-year-old white female presents with compl...	Allergy / Immunology	Allergic Rhinitis	SUBJECTIVE: This 23-year-old white female pr...	allergy / immunology, allergic rhinitis, aller...
1	Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult - 2	PAST MEDICAL HISTORY: He has difficulty climb...	bariatrics, laparoscopic gastric bypass, weigh...
2	Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult - 1	HISTORY OF PRESENT ILLNESS: I have seen ABC ...	bariatrics, laparoscopic gastric bypass, heart...
3	2-D M-Mode. Doppler.	Cardiovascular / Pulmonary	2-D Echocardiogram - 1	2-D M-MODE: , 1. Left atrial enlargement wit...	cardiovascular / pulmonary, 2-d m-mode, dopple...
4	2-D Echocardiogram	Cardiovascular / Pulmonary	2-D Echocardiogram - 2	1. The left ventricular cavity size and wall ...	cardiovascular / pulmonary, 2-d, doppler, echo...

- iii. Detect Abbreviation for the entered healthcare data

```
!python -m pip install --user spacy
!python -m spacy download en_core_web_sm

import en_core_web_sm
nlp = en_core_web_sm.load()

# This function generate anotation for each entities and label
def generate_annotation(texts):
    annotations = []
    for text in texts:
        doc = nlp(text)
        entities = []
        for ent in doc.ents:
            entities.append((ent.start_char, ent.end_char, ent.label_))
        annotations.append((text, {'entities': entities}))
    return annotations

# Extract text entities and labels from the dataset (transcription)
raw_df['transcription'] = raw_df['transcription'].astype(str)
medical_doc = raw_df['transcription'].tolist()

# Let's generate annotations
annotations = generate_annotation(medical_doc)

# Let's print documents and annotations
print("Document:")
print(annotations[0][0]) # first document text
print("Annotations:")
print(annotations[0][1]) # annotation for the first document
```

Document:
SUBJECTIVE: This 23-year-old white female presents with complaint of allergies. She used to have allergies when she lived in Seattle but she thinks they are worse here. In the past, she has tried Zyrtec, Allegra, last summer, two weeks ago, daily, Ortho Tri-Cyclen, and the Allegra, ALLERGIES. She has no known medicine allergies, OBJECTIVE: Vitals: Weight was 130 pounds and blood pressure 124/78, HEENT: Her throat was mildly erythematous without exudate. Nasal mucosa was erythematous and swollen. Only clear drainage was seen. TMs were clear, Neck: Supple without adenopathy, Lungs: Clear, ASSESSMENT: Allergic rhinitis, PLAN: 1. She will try Zyrtec instead of Allegra again. Another option will be to use loratadine. She does not think she has prescription coverage so that might be cheaper. 2. Samples of Nasonex two sprays in each nostril given for three weeks. A prescription was written as well.

iv. Link to UMLS and others with the Entity Linker.

```
import en_core_web_sm
nlp = en_core_web_sm.load()

# Create distinct colours for labels

col_dict = {}
s_colours = ['#e6194b', '#3cb44b', '#ffe119', '#ffdb7f', '#f58220', '#f032e6', '#42d4f4']
for label, colour in zip(nlp.pipe_labels['ner'], s_colours):
    col_dict[label] = colour

options = {'ents': nlp.pipe_labels['ner'], 'colors': col_dict}

transcription = raw_df['transcription'][0]
doc = nlp(transcription)

spacy.displacy.render(doc, style = 'ent', jupyter = True, options = options)

[(ent.text, ent.label_) for ent in doc.ents]
```

SUBJECTIVE: This 23-year-old white female presents with complaint of allergies. She used to have allergies when she lived in Seattle but she thinks they are worse here. In the past, she has tried Claritin, and Zyrtec. Both worked for short time but then seemed to lose effectiveness. She has used Allegra also. She used that last summer and she began using it again two weeks ago. It does not appear to be working very well. She has used over-the-counter sprays but no prescription nasal sprays. She does have asthma but does not require daily medication for this and does not think it is flaring up. MEDICATIONS: Her only medication currently is Ortho Tri-Cyclen and the Allegra. ALLERGIES: She has no known medicine allergies, OBJECTIVE: Vitals: Weight was 130 pounds and blood pressure 124/78, HEENT: Her throat was mildly erythematous without exudate. Nasal mucosa was erythematous and swollen. Only clear drainage was seen. TMs were clear, Neck: Supple without adenopathy, Lungs: Clear, ASSESSMENT: Allergic rhinitis, PLAN: 1. She will try Zyrtec instead of Allegra again. Another option will be to use loratadine. She does not think she has prescription coverage so that might be cheaper. 2. Samples of Nasonex two sprays in each nostril given for three weeks. A prescription was written as well.

```
[('23-year-old', 'DATE'),
 ('Seattle', 'GPE'),
 ('Claritin', 'PERSON'),
 ('Zyrtec', 'LANGUAGE'),
 ('Allegra', 'ORG'),
 ('last summer', 'DATE'),
 ('two weeks ago', 'DATE'),
 ('daily', 'DATE'),
 ('Ortho Tri-Cyclen', 'PERSON'),
 ('Allegra', 'ORG'),
 ('130 pounds', 'QUANTITY'),
 ('Nasal', 'ORG'),
 ('Allergic', 'ORG'),
 ('Zyrtec', 'NORP'),
 ('Allegra', 'ORG'),
 ('Samples of Nasonex', 'ORG'),
 ('two', 'CARDINAL'),
 ('three weeks', 'DATE')]
```

6. Post-Experiments Exercise

A. Extended Theory:

a. List and explain other NLP Libraries in python used for Healthcare applications.

B. Conclusion:

i. Write what was performed in the program (s) .

ii. What is the significance of program and what Objective is achieved?

C. References:

[1]<https://www.andrewvillazon.com/clinical-natural-language-processing-python/>

[2]https://colab.research.google.com/drive/1SYRRD7w7XEz2s6O4pv0pQMtFjcUOVmKI?usp=sharing&scrollTo=_z5GuCGNKPxK